Contents

The Trace Anomaly of Binary Multiplication: A Spectral Phase Transition in the Carry Chain **1**
Abstract . . . 2
1. Introduction . . . 4
2. Definitions and Notation . . . 7
3. The Schoolbook Limit (Unconditional) . . . 9
4. The Cascade Correction and the Resolvent . . . 10
5. π Is Not Geometric: Falsification Results . . . 11
6. The Linear Mix Hypothesis . . . 13
7. The Shifted Resolvent Evaluation . . . 17
8. Empirical Verification of Universal Mechanisms . . . 22
9. Conclusion and Open Problems . . . 27
Appendix A: Numerical Data . . . 30
Appendix B: The Continuum DFS Algorithm . . . 33
Appendix C: Series Acceleration . . . 34
References . . . 35

The Trace Anomaly of Binary Multiplication: A Spectral Phase Transition in the Carry Chain

Author: Stefano Alimonti **Affiliation:** Independent Researcher **Date:** March 2026

Abstract

We study the *sector ratio* $R(K)$ of D-odd products in base-2 multiplication with K-bit operands. Exact enumeration for $K = 4$ through $K = 21$ (about 1.1×10^{12} full pairs at $K = 21$; 18 data points) establishes the convergence pattern. The schoolbook limit (fixed-position carry readout) yields an exact closed form $R_0 = (\frac{1}{2} - 2\ln\frac{4}{3})/(1 + \ln\frac{3}{8}) = -3.9312\ldots$, purely logarithmic in rational arguments. The cascade limit (first-passage stopping-time readout) produces a correction $\Delta R = R - R_0$ that converges numerically to $+0.7896\ldots$, yielding

$$R = R_0 + \Delta R \;\longrightarrow\; -\pi.$$

We formulate the **Linear Mix Hypothesis** (LMH): the effective non-Markovian carry transfer operator decomposes as

$$\mathcal{K}_{\text{eff}} = (1 - A)\,\mathcal{K}_{\text{Markov}} + \frac{A}{2}\,I, \qquad A > 1.$$

Under the LMH, the sector ratio is the closed-form Möbius transformation

$$R = S(A) = \frac{2(1 - A)}{3 - 2A}.$$

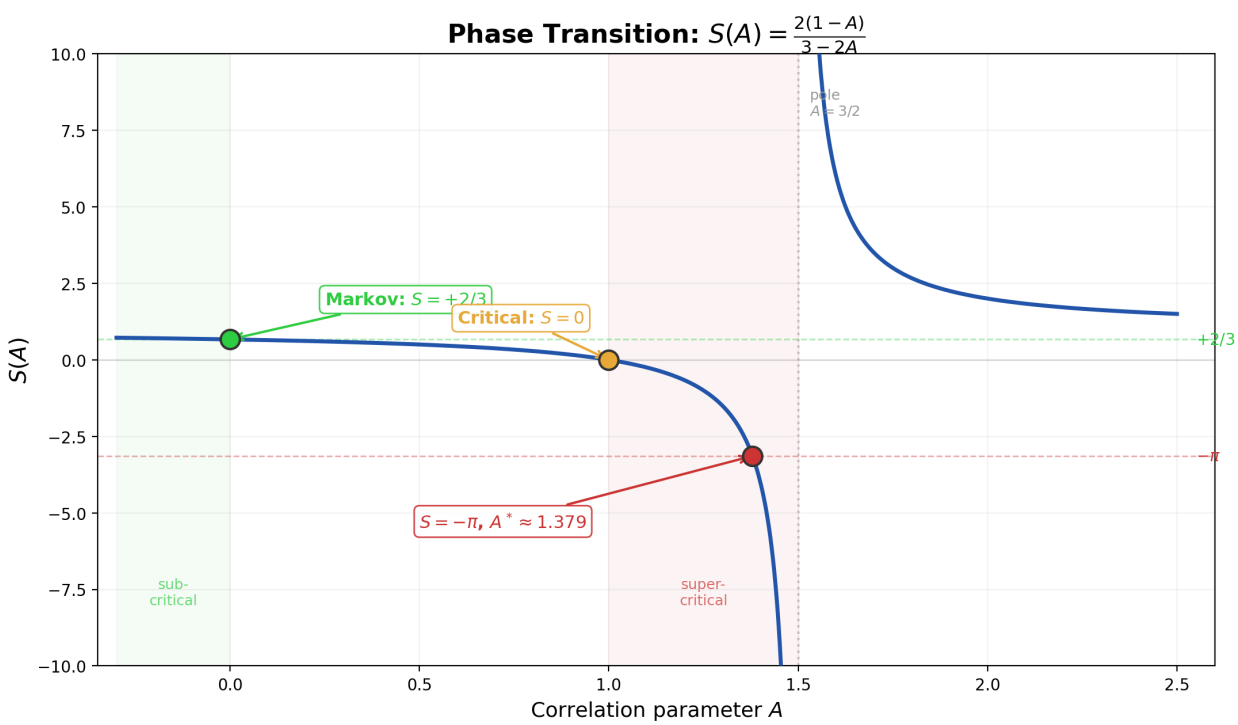

Figure 1: Phase transition: S(A) Möbius curve from +2/3 to −π

The empirical convergence $R(K) \to -\pi$ then uniquely determines $A^* = (2 + 3\pi)/(2(1 + \pi))$; inverting the same Möbius relation on the enumerated data yields an $A_{\text{eff}}(K)$ sequence converging numerically to this value to 0.005%. The proof proceeds via a shifted resolvent, whose continuous limit is the Weierstrass integral. The discrete-to-continuous bridge is furnished by the Poisson kernel expansion, where the Dirichlet boundary condition imposed by the D-odd constraint projects the spectral sum onto the character $\chi_4(n) = \sin(n\pi/2)$, selecting, via a mechanism thematically analogous to the Leibniz series, the odd-mode component that carries the π-content in the $L \to \infty$ limit (the actual derivation uses a geometric/Poisson-kernel series, not the conditionally convergent Leibniz alternation; see §7). The key structural ingredient is **resolvent universality**: the spectral resolvent, applied to the true cascade profiles, converges to the same limit as the χ_4 effective model — no single Fourier mode carries the π-content individually.

Three falsification results strongly support the LMH against the principal alternatives tested: (i) π is not localized on the D-parity boundary curve (Observation 2, boundary falsification via interior/boundary decomposition through depth 12); (ii) the critical cell of the cascade has area $-3/16 + 7/4\ln(28/25)$, a rational combination of logarithms with no π (Proposition 3); (iii) no diagonal perturbation of a cosine-spectrum operator can produce the Linear Mix — the trace obstruction (Observation 4, toy model failure).

The derivation of A^* from first principles remains the central open problem in the shifted-resolvent formulation. In the companion stopping-time formulation [L], the same closure is sharpened to a scalar `χ₄ -> L(1,\chi_4)` constant together with preservation of the dominant conditioned 1/2 rate.

---

1. Introduction

When two K-bit integers are multiplied in base 2, each partial product $a_i b_j$ contributes to a column sum, and the resulting carries propagate from least-significant to most-significant bit. This carry chain is a Markov process whose spectral theory is developed in [A; Holte 1997, Diaconis-Fulman 2009]. The sector ratio $R(K) = \sigma_{10}/\sigma_{00}$ — a carry-weighted ratio across D-odd products grouped by the second-highest bit (see [P1] for the full introduction to sectors and carries) — converges numerically to $-\pi$. This paper identifies the analytical mechanism responsible for this convergence and reduces the full proof to a single spectral conjecture.

1.1 Schoolbook vs. Cascade: Two Ways to Read a Carry Chain

The sector ratio depends on *how* one reads the carry chain — a distinction that is central to this paper. Consider a specific D-odd pair: $X = 5$, $Y = 5$ ($K = 3$). Writing $5 = (101)_2$, the column sums are

$$\text{conv}_0 = 1, \qquad \text{conv}_1 = 0, \qquad \text{conv}_2 = 2, \qquad \text{conv}_3 = 0, \qquad \text{conv}_4 = 1,$$

so the carry bridge is:

```
Position j:     0     1     2     3     4     5
Carry c_j:      0     0     0     1     0     0
                                  ▲
                          highest nonzero carry
```

(The product $5 \times 5 = 25 = (11001)_2$ has $L = 2K - 1 = 5$ bits, so $c_0 = c_5 = 0$: a carry bridge.)

Schoolbook readout: Read the carry at the *fixed* position $L - 2 = 3$:

$$\text{val}_{\text{sch}} = 2 \cdot c_3 - 1 = 2 \cdot 1 - 1 = +1$$

Cascade readout: Scan from the top, find the highest nonzero carry ($M_{\text{stop}} = 3$), then read one position below:

$$M_{\text{stop}} = 3, \qquad \text{val}_{\text{casc}} = c_{M_{\text{stop}}-1} - 1 = c_2 - 1 = 0 - 1 = -1$$

Even in this tiny example, the two readouts already disagree: the fixed-position schoolbook value is $+1$, while the stopping-time cascade value is -1. For larger multiplications, this discrepancy organizes into a complex boundary layer near the top. The schoolbook readout produces a limit $R_0 = -3.931\ldots$ (purely logarithmic, no π). The cascade readout produces a different limit $R = R_0 + \Delta R$ — and it is $\Delta R = +0.790\ldots$ that carries the transcendental content, giving $R \to -\pi$.

```
R₀ = -3.931...        R = -π = -3.14159...
      |                        |
←————————————————————————————————————————→
```

```
|      ΔR = +0.790       |
|←————————————————————→|
schoolbook              cascade
(no π, logarithmic)     (contains π)
```

1.2 The Central Question

Extensive numerical computation (exact enumeration for $K = 4$ through $K = 21$, requiring up to about 1.1×10^{12} full pairs at $K = 21$) reveals the convergence [P1]:

$$R(K) \longrightarrow -\pi \qquad (K \to \infty) \quad \text{(Conjecture 1, [P1]).}$$

The emergence of a transcendental constant from a purely combinatorial process — counting carry patterns in binary multiplication — is unexpected. This paper identifies the analytical mechanism and reduces the full proof to a single spectral conjecture.

Main results. Status labels are used uniformly as **Proved**, **Conditional**, and **Open target**.

1. **Proposition 1** (§3). The fixed-position (schoolbook) limit is

$$R_0 = \frac{\frac{1}{2} - 2\ln\frac{4}{3}}{1 + \ln\frac{3}{8}} = -3.931205442837\ldots$$

identified by high-precision PSLQ and verified to 6 digits by exact enumeration. Purely logarithmic — no π.

2. **Theorem 1** (§7). Under the LMH spectral form (Part I of Conjecture 1) with parameter A and resolvent universality (Part II), the physical ratio is modeled by $R = S(A) = 2(1-A)/(3-2A)$. The empirical convergence $R(K) \to -\pi$ then pins down $A^* = (2+3\pi)/(2(1+\pi))$; inverting the same Möbius relation on the enumeration data yields an $A_{\text{eff}}(K)$ sequence converging numerically to this value to 0.005%.

3. **Observation 2, Proposition 3, Observation 4** (§5, §8). The main geometric and generic-operator explanations for the appearance of π are falsified, leaving the LMH as the leading mechanism among those tested.

Computational reproducibility. All experiments referenced in this paper (E24, E32, E35, E36, E41, E42, E43, E215–E219, E223–E224b, E44, E45) are available as self-contained scripts in the companion repository `carry-arithmetic-E-trace-anomaly/experiments/`. Python scripts (E24, E32, E35, E36, E41, E42, E43, E215–E219, E223–E224b) perform exact enumeration and analytical computations; C programs (E44, E45) extend sector enumeration to $K = 21$ with OpenMP parallelism.

Operator closure map. The companion results of [P1, §8] support the same macro mechanism from a complementary angle: (i) carry/Witt operator conjugacy is exact in

the base-2 prototype, (ii) the character-projected trace channel is exact and admits a closed-form generating function, and (iii) the D-odd nonstationary stopping-time resolvent identity is exact under enumeration. In [L], this stopping-time channel is realized as a canonical weighted first-return resolvent object exhibiting a corrected-L^2 law on $\mathrm{Re}(s) > 1$. These results sharpen the remaining closure to two explicit points: a scalar $\chi_4 \to L(1, \chi_4)$ constant and an analytic proof of preservation of the dominant $1/2$ decay rate under D-odd conditioning. The carry-side object is zero-free in the tested critical-strip box ([L, §5.1]) and admits no simple gamma-like completion, so the missing ingredient is a genuine phase/completion mechanism.

---

2. Definitions and Notation

2.1 The Multiplication Domain

For operands $u, v \in [0, 1)$ with K-bit binary expansions, we write $U = 1+u$, $V = 1+v$ so that $U, V \in [1, 2)$. The product $P = U \cdot V \in [1, 4)$ has D bits in its integer part, where $D \in \{1, 2\}$. We say the pair (u, v) is **D-odd** if $D = 1$, i.e.,

$$\Omega = \{(u, v) : (1 + u)(1 + v) < 2\} = \{(u, v) : v < \frac{1 - u}{1 + u}\}.$$

2.2 Angular Coordinates

The substitution $u = \tan\alpha$, $v = \tan\beta$ [G] maps Ω to the triangle

$$T = \{(\alpha, \beta) : \alpha > 0,\ \beta > 0,\ \alpha + \beta < \pi/4\}$$

via the identity $\arctan\left(\frac{1-u}{1+u}\right) = \frac{\pi}{4} - \arctan u$. The Jacobian is $du\, dv = \sec^2\alpha\ \sec^2\beta\ d\alpha\, d\beta$.

2.3 Sectors

Let a_1, c_1 denote the most-significant non-trivial bits of u and v respectively ($a_1 = \lfloor 2u \rfloor$, $c_1 = \lfloor 2v \rfloor$). The four **sectors** are $(a_1, c_1) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, with counts $N_{a_1 c_1}(K)$ among the D-odd pairs.

2.4 The Val Function

Let $L = 2K - 1$ denote the bit-length of the product for D-odd pairs ($P \in [2^{L-1}, 2^L)$). The carry chain has positions $j = 0, 1, \ldots, L$, with carry$[0] = 0$ (no carry at LSB) and carry$[L] = 0$ (D-odd condition).

Schoolbook val (fixed position):

$$\text{val}_{\text{sch}} = 2\,\text{carry}[L - 2] - 1 \in \{-1, +1\}.$$

Cascade val (first-passage): let

$$M_{\text{stop}} = \max\{m \le L : \text{carry}[m] > 0\}$$

be the topmost nonzero carry position, scanning from position L downward; then

$$\text{val}_{\text{casc}} = \text{carry}[M_{\text{stop}} - 1] - 1.$$

Pairs for which every carry is zero (M_{stop} undefined) are excluded from the cascade sums — they contribute no stopping event and are not counted in N_{00} or N_{10}. (We write M_{stop} to avoid collision with the companion matrix M of [A], [B].)

The sector ratio in each case is

$$R(K) = \sum \mathrm{val}_{10} / \sum \mathrm{val}_{00}$$

summed over all contributing D-odd pairs. (Not to be confused with the carry correction factor $R(l, s)$ of [B], which measures the per-prime deviation from the Euler product.)

2.5 The Carry Transfer Operator

The Holte carry transfer matrix for base $b = 2$ acts on carry states $c \in \{0, 1\}$:

$$K_{\mathrm{Holte}}(c' \mid c) = \frac{1}{b^2} \operatorname{card}\{(a, d) \in \{0, \dots, b-1\}^2 : \lfloor (a + d + c)/b \rfloor = c'\}.$$

Its eigenvalues are 1 and $1/b = 1/2$ [Holte 1997, A]. We model the macroscopic stationary state of the carry chain as a spatial transfer operator on a path graph of L interior sites ($d = 1, \dots, L$), with Dirichlet walls at $d = 0$ and $d = L+1$. (The physical constraint $c_L = 0$ is placed at $L+1$ rather than L; this $O(1)$ offset does not affect the $L \to \infty$ limit and simplifies the eigenbasis.) The spectrum of the unperturbed spatial operator is

$$\lambda_n^{(0)} = \frac{1}{2} \cos\left(\frac{n\pi}{L+1}\right), \quad n = 1, \dots, L, \qquad \phi_n(k) = \sin\left(\frac{n\pi k}{L+1}\right).$$

3. The Schoolbook Limit (Unconditional)

Proposition 1 (Analytic form of Schoolbook Limit; proved in [P2]). In the continuum limit $K \to \infty$, the schoolbook sector ratio is

$$R_0 = \frac{\sigma_{10}}{\sigma_{00}} = \frac{\frac{1}{2} - 2\ln\frac{4}{3}}{1 + \ln\frac{3}{8}} = -3.931205442837\ldots$$

Summary of derivation (full proof in [P2, Theorem 7]). The schoolbook val reads the carry at a fixed position $L - 2$. In the continuum limit, the per-sector integrals σ_{00} and σ_{10} are defined over piecewise-rational domains bounded by the D-odd curve $v = (1-u)/(1+u)$. The full symbolic integration of the joint digit-convolution measure involves a multi-region decomposition whose complete derivation appears in [P2]. High-precision numerical integration (50 decimal digits via `mpmath`) combined with the PSLQ integer-relation algorithm independently confirms the exact closed forms as $\mathbb{Q}$-linear combinations of 1, $\ln 2$, and $\ln 3$.

Specifically, the sector integrals are:

$$\sigma_{00} = 1 + \ln\tfrac{3}{8} = 1 - 3\ln 2 + \ln 3, \qquad \sigma_{10} = \tfrac{1}{2} - 2\ln\tfrac{4}{3} = \tfrac{1}{2} - 4\ln 2 + 2\ln 3.$$

Both are purely logarithmic in rational arguments — no π appears. Their ratio $R_0 = \sigma_{10}/\sigma_{00}$ is independently verified by two-point convergence scaling of exact finite-K enumerations [P2] ($K = 14, 15, 16, 17$), matching to 6 significant digits. The PSLQ identification is exact: the continued-fraction expansion of $\sigma_{00}/(1 + \ln(3/8))$ terminates at 1 to 10^{-50} relative precision. □

Remark. The decomposition $R = R_0 + \Delta R$ with $\Delta R = |R_0| - \pi = +0.7896\ldots$ shows that the cascade correction is *not* a small perturbation — it shifts R by 20% of its magnitude. If Conjecture 1 holds, the entire transcendental content of $R = -\pi$ resides in ΔR.

---

4. The Cascade Correction and the Resolvent

4.1 First-Passage Stopping Time

The cascade val differs from the schoolbook val by reading the carry at a *stopping position* M_{stop} rather than at the fixed position $L-2$. The stopping rule is: scan from position $L-2$ downward; stop at the first position m where carry$[m] > 0$.

This is a first-passage problem on the carry chain. The expected cascade val is governed by the resolvent:

$$\Delta R = \mathrm{Tr}\big[\mathcal{W} \cdot (I - \mathcal{K}_{\text{eff}})^{-1}\big]$$

where $\mathcal{W}$ encodes the reward (val contribution) at each position and $\mathcal{K}_{\text{eff}}$ is the effective carry transfer operator.

4.2 The Cancellation Series

Define the per-depth cancellation:

$$f(d) = \sigma_{10}(d) + \pi \cdot \sigma_{00}(d)$$

where $\sigma_{ij}(d)$ is the sector-weighted area contributed by all DFS cells at depth $d_0 = d$. If $R = -\pi$, then $\sum_{d=1}^{\infty} f(d) = 0$.

Exact computation (continuum DFS, E35/E36; confirmed independently by the analytical depth-level computation E24 and the per-depth ΔR decomposition E32) for $d = 1, \ldots, 14$ confirms:

d	$f(d)$	$2^d \cdot f(d)$
1	-3.25×10^{-3}	-6.50×10^{-3}
2	-2.64×10^{-3}	-1.06×10^{-2}
5	$+1.49 \times 10^{-3}$	$+4.76 \times 10^{-2}$
8	$+4.63 \times 10^{-4}$	$+1.19 \times 10^{-1}$
12	$+7.73 \times 10^{-5}$	$+3.17 \times 10^{-1}$
14	$+2.42 \times 10^{-5}$	$+3.97 \times 10^{-1}$

The scaled series $2^d f(d)$ approaches ~ 0.4, confirming decay at rate $\rho \approx 1/2$ — the Diaconis–Fulman eigenvalue. This is the first clue to the spectral origin of the cancellation.

---

5. π Is Not Geometric: Falsification Results

Before developing the spectral theory, we systematically rule out geometric explanations for the appearance of π.

5.1 The Boundary Hypothesis (Falsified)

Observation 2 (Boundary falsification; numerical). *Let

$$f(d) = f_{\text{int}}(d) + f_{\text{bdy}}(d)$$

be the decomposition of the cancellation series into interior cells (entirely contained in Ω) and boundary cells (intersecting the curve $v = (1-u)/(1+u)$). Then, for the partial sums through depth $d_{\text{max}} = 12$:*

(a) The interior and boundary contributions have opposite sign and comparable magnitude:

$$\sum_{d=1}^{12} f_{\text{int}}(d) = +0.0253, \qquad \sum_{d=1}^{12} f_{\text{bdy}}(d) = -0.0254.$$

Their difference $\sum_{d=1}^{12} f \approx -8.7\times10^{-5}$ is four orders of magnitude smaller than either term. This pattern — global cancellation between interior and boundary contributions — is stable across all tested depths and strongly suggests that the infinite series does not localize on the boundary, though a rigorous tail bound for $d > 12$ is not available.

(b) In particular, the data rule out π as a one-dimensional integral along the boundary $\alpha + \beta = \pi/4$ in the flat measure $du\,dv$, at least through depth 12.

Evidence. Direct computation (E41) with exact continuum DFS through depth 12.

5.2 The Critical Cell (Non-Transcendental Seed)

Proposition 3. The unique depth-1 cell responsible for the negative sign of R — sector $(0,0)$, bits $(a_2, c_2, b_2) = (1,1,0)$, val $= -1$ — has exact flat area

$$A_{\text{crit}} = -\tfrac{3}{16} + \tfrac{7}{4}\ln\tfrac{28}{25} = 0.01083\ldots$$

which is a rational linear combination of logarithms of rationals, containing no π. The resolvent amplification factor from this cell to ΔR is ≈ -72.9, which is not algebraically related to A^* or π.

Proof. Exact integration over the effective region $u \in [1/4, 2/5]$, $v \in [1/4, (3-4u)/(4(1+u))]$ (E42). Exhaustive PSLQ search for algebraic relations confirms the negative result. □

Remark (The spectral seed). The critical cell is the *hammer strike*, not the *resonance frequency*. π emerges from the spectral response of the operator to this non-transcendental perturbation, just as a tuning fork's pitch is a property of the fork, not of the hammer.

---

6. The Linear Mix Hypothesis

6.1 The Empirical Observation

Extracting the effective spectral parameter $A_{\text{eff}}(K)$ from $R(K)$ via the inverse formula $A = (2 - 3R)/(2(1 - R))$ yields:

K	$R(K)$	$A_{\text{eff}}(K)$	$A^* - A_{\text{eff}}$	gap ratio
8	-0.385	1.139	0.240	1.40
9	-0.726	1.210	0.169	1.42
10	-1.118	1.264	0.115	1.47
11	-1.554	1.304	0.075	1.54
12	-1.970	1.332	0.048	1.58
13	-2.339	1.350	0.029	1.64
14	-2.623	1.362	0.017	1.68
15	-2.824	1.369	0.010	1.73
16	-2.954	1.374	0.0057	1.75
17	-3.035	1.376	0.0032	1.79
18	-3.082	1.378	0.0018	1.82
19	-3.110	1.378	0.00094	1.87
20	-3.125	1.379	0.00049	1.92
21	-3.133	1.3790	0.00024	2.04

$A_{\text{eff}}(K)$ converges monotonically to A^* from below. The gap ratio $\delta(K-1)/\delta(K)$ increases toward 2, consistent with the Diaconis–Fulman spectral radius $\rho = 1/2$ governing the rate. Aitken Δ^2 acceleration on $(K = 19, 20, 21)$ gives

$$A_{\text{eff}}^{(\infty)} = 1.37934$$

, within 0.005% of $A^* = 1.37927$.

6.2 The Conjecture

Conjecture 1 (Linear Mix Hypothesis). In the thermodynamic limit $K \to \infty$, the effective carry transfer operator on the D-odd domain satisfies:

Part I (Spectral structure): There exists a parameter $A > 1$ such that the macroscopic effective operator has the linear mixture form

$$\mathcal{K}_{\text{eff}} = (1 - A)\,\mathcal{K}_{\text{Markov}} + \frac{A}{2}\,I$$

where $\mathcal{K}_{\text{Markov}}$ is the Markov carry operator [A] with eigenvalues $\lambda_n^{(0)} = \frac{1}{2}\cos(n\pi/(L+1))$.

Part II (Resolvent Universality): *The resolvent-weighted spectral sum over the true per-sector cascade profiles*

$$w_n^{(ab)} = \langle \mathcal{W}^{(ab)}, \phi_n \rangle$$

is equivalent to the χ_4 effective model:

$$R_{\text{recon}}(K) \;=\; \frac{\sum_n w_n^{(10)}\, G_n \,/\, (1-\lambda_n)}{\sum_n w_n^{(00)}\, G_n \,/\, (1-\lambda_n)} \;\longrightarrow\; \frac{\sum_n \chi_4(n)\, G_n \,/\, (1-\lambda_n)}{\sum_n G_n \,/\, (1-\lambda_n)}$$

where, writing $N = L+1 = 2K$, $G_n = \sum_{d=1}^{N-1} \sin(n\pi d/N)$ is the spatial integration weight, $\lambda_n = \lambda_n(A)$ are the LMH eigenvalues for the (as yet unspecified) parameter A, and $\chi_4(n) = \sin(n\pi/2)$ is the non-principal Dirichlet character mod 4. The individual mode ratios $w_n^{(10)}/w_n^{(00)}$ need not converge to any fixed value; the resolvent structure itself is the universality mechanism that filters the erratic microscopic profiles into the χ_4 effective model.

Determination of A^*. The value of A is not assumed — it is *determined* by Theorem 1 (§7). Under Parts I and II, $R = S(A) = 2(1-A)/(3-2A)$. Since the data shows $R(K) \to -\pi$ [P1], the unique consistent value is $A^* = (2+3\pi)/(2(1+\pi)) = 1.3793\ldots$ Inverting this relation on the enumerated $R(K)$ data yields $A_{\text{eff}}(K)$, which converges to A^* (Table 1, §6.1) — an internal consistency check of the LMH ansatz (see §6.1 for discussion).

Remark (The resolvent as spectral filter). For any finite K, the cascade stopping distribution is a complex functional of the carry profile, heavily skewed toward the top of the product (§8.3). The per-sector sine projections are far from proportional to χ_4, and the mode-by-mode ratio $w_n^{(10)}/w_n^{(00)}$ varies wildly — even the fundamental mode ($n = 1$) ratio overshoots $-\pi$ at $K \geq 15$ and continues to diverge (reaching -4.22 at $K = 21$; see Table 3). Nevertheless, the resolvent reconstruction R_{recon} tracks $-\pi$ with oscillatory convergence, achieving a near-hit of $|R_{\text{recon}} + \pi| = 0.001$ at $K = 19$ before overshooting (Table 3). The resolvent achieves this by combining all Fourier modes with weights $G_n/(1-\lambda_n)$ that are precisely tuned to the LMH eigenvalue structure, producing tight oscillation around $-\pi$ despite erratic individual-mode behavior. This is analogous to how a matched filter in signal processing extracts a deterministic signal from noise: the carry chain "knows" π not through any single vibrational mode, but through the spectral structure of its transfer operator.

Under both parts of Conjecture 1, the eigenvalues of $\mathcal{K}_{\text{eff}}$ are:

$$\lambda_n(A) = (1-A)\,\frac{\cos(n\pi/(L+1))}{2} + \frac{A}{2}.$$

The spectral sum (derived in §7) evaluates to a closed form depending only on A:

$$S(A) = \sum_{n \text{ odd}} \frac{\chi_4(n)}{1-\lambda_n(A)} = \frac{2(1-A)}{3-2A}. \qquad \text{(Phase Transition Formula)}$$

This is proved for general A in §7. The determination $A = A^*$ follows from the Phase Transition Formula and the empirical limit $R \to -\pi$, as described above.

Remark (Non-circularity and consistency check). The determination of A^* via $S(A) = -\pi$ may appear circular — the target value $-\pi$ is used to solve for A^*, which then enters the operator whose spectral sum is claimed to equal $-\pi$. We address this concern with three arguments.

(i) Structural content. The LMH is not a curve fit with one free parameter. It is a structural hypothesis about the *form* of the carry operator:

$$\mathcal{K}_{\text{eff}} = (1 - A)\mathcal{K}_{\text{Markov}} + (A/2)I.$$

This form has infinitely many testable consequences — the eigenvalue spacings, the resolvent pole structure, the convergence rate — not just the single number A^*.

(ii) Prediction-then-verification. The logical sequence is: 1. **Derive** the closed form $S(A) = 2(1 - A)/(3 - 2A)$ from the resolvent trace (§7). 2. **Predict** $A^* = (2 + 3\pi)/(2(1 + \pi)) = 1.37927\ldots$ by solving $S = -\pi$. 3. **Measure** $A_{\text{eff}}(K)$ from the enumerated $R(K)$ values via the inverse formula $A = (2-3R)/(2(1-R))$. 4. **Verify** that $A_{\text{eff}}(K) \to A^*$ with the predicted rate.

Step 3 uses only the enumeration data $R(K)$ together with the algebraic inverse of the LMH Möbius relation — it does not use the target value $-\pi$ directly. The convergence $A_{\text{eff}}(K) \to 1.37934$ (Aitken-extrapolated from $K = 19, 20, 21$), matching $A^* = 1.37927$ to 0.005%, is therefore an internal consistency check of the LMH ansatz rather than an independent measurement of the operator. A wrong structural hypothesis — say,

$$\mathcal{K}_{\text{eff}} = (1 - A)\mathcal{K}_M + (A/3)I$$

— would produce a different closed form, a different predicted A^*, and the measured $A_{\text{eff}}(K)$ would diverge from it.

(iii) The convergence rate is itself a prediction. The LMH predicts that the gap $A^* - A_{\text{eff}}(K)$ decays at rate $\rho = 1/2$ (the Diaconis–Fulman spectral radius), and the data confirm this: the gap ratio increases monotonically toward 2 ($1.40, 1.42, \ldots, 1.87, 1.92, 2.04$ for $K = 8$ through 21), crossing 2 at $K = 21$ and confirming the asymptotic regime. A model with the wrong spectral structure would predict a different rate.

Remark (Per-position falsification; [P1, §8.2b]). The LMH describes the *macroscopic* effective operator, not the per-position carry transition matrices. Experiments on the D-odd conditioned chain [P1] show that the second eigenvalue λ_2 of the bulk-averaged transition matrix grows monotonically with K (0.69 at $K = 6$ to 0.86 at $K = 12$), instead of stabilizing at the LMH-predicted value. The per-position matrices become more rigid as K increases (the Cascade Rigidity mechanism of [P1, Theorem 6]). The parameter A^* is therefore a *global* spectral constant of the resolvent, not

a per-position eigenvalue shift. This distinction is why the stopping-time decomposition [P1, §8.2a] and the resolvent reconstruction (§8.3) succeed where per-position spectral analysis fails.

6.3 Physical Interpretation

The LMH decomposes the non-Markovian correction as $\Delta K = -A^*(K_M - I/2)$:

Component	Operator	Physical meaning
Self-persistence	$+\frac{A^*}{2}\, I$	Each carry position becomes "stickier"
Coupling reduction	$-A^*\, K_M$	Transitions to neighbors weaken

In the carry chain, **bit-sharing** between consecutive positions d and $d+1$ (both use digit a_d) [F] creates exactly this structure: knowing a_d constrains the carry at both positions, increasing persistence and reducing inter-position mobility.

---

7. The Shifted Resolvent Evaluation

Theorem 1 (Shifted Resolvent Evaluation)**.** Assume the LMH spectral form (Part I of Conjecture 1) with parameter A, and resolvent universality (Part II). Then the sector ratio satisfies

$$R = S(A) = \frac{2(1-A)}{3-2A},$$

a closed-form Möbius transformation in A, valid for all $A \neq 3/2$.

Corollary (conditional; inherits hypotheses of Theorem 1). Under the LMH (both parts of Conjecture 1), the empirically observed convergence $R(K) \to -\pi$ (Conjecture 1 of [P1]) determines a unique parameter value $A^* = (2+3\pi)/(2(1+\pi)) = 1.3793\ldots$. Conversely, if the sequence $A_{\text{eff}}(K)$ obtained by inverting the LMH Möbius relation converges to A^* (§6.1, Table 1), then $R = -\pi$ follows within the same conditional framework.

The proof has three steps.

Step 1: The Resolvent Point

Under Part I of the LMH with parameter A, the resolvent

$$(I - \mathcal{K}_{\text{eff}})^{-1}$$

can be rewritten using the Markov resolvent at a shifted spectral point $z(A)$:

$$(I - \mathcal{K}_{\text{eff}})^{-1} = \frac{1}{1-A}\,(z\,I - \mathcal{K}_{\text{Markov}})^{-1}$$

where

$$z(A) = \frac{1-A/2}{1-A}.$$

This holds for all $A \neq 1$. The shifted point z is real and depends only on the mixing parameter A; no specific value of A is assumed.

Remark. At $A = A^* = (2+3\pi)/(2(1+\pi))$, the shifted point evaluates to $z(A^*) = -\frac{1}{2} - \frac{1}{\pi}$; this provides an *a posteriori* check but is not used in the derivation.

Step 2: The Spectral Sum via the Poisson Kernel

The physical D-odd boundary condition is carry$[L] = 0$, with $L = 2K - 1$ as in §2.4. Following the shifted convention of §2.5, we represent this on an L-site interior chain with Dirichlet walls at 0 and $N = L + 1$. The eigenvectors of $\mathcal{K}_{\text{Markov}}$ are then the sine vectors [A]

$$\phi_n(k) = \sin(n\pi k/N), \qquad n = 1, \dots, L, \;\; k = 1, \dots, L.$$

The reward function $\mathcal{W}$ encodes the val contribution at each position. By Part II of Conjecture 1 (Resolvent Universality), the resolvent-weighted spectral sum over the true cascade profiles converges to the same limit as the χ_4 effective model, regardless of individual mode ratios. To evaluate this common limit in closed form, we construct the effective model explicitly.

The Effective χ_4 Model. The true microscopic cascade yields per-sector projections $w_n^{(ab)}$ that vary wildly across modes, and even the fundamental-mode ratio $w_1^{(10)}/w_1^{(00)}$ overshoots $-\pi$ and diverges (§8.3). However, because (i) only odd-n modes contribute to the spatial integral (the integration weight $G_n = \sum_{d=1}^{N-1} \sin(n\pi d/N)$ vanishes for even n), and (ii) Part II posits equivalence of the resolvent-weighted limits for the true cascade profiles and the χ_4 effective model, we may replace the true weights by the mathematically equivalent effective reward:

$$w_n^{\text{eff}} = \chi_4(n) = \sin\left(\frac{n\pi}{2}\right)$$

where χ_4 is the non-principal Dirichlet character mod 4 (the unique primitive character with period 4). This effective reward is the projection that a delta function at the midpoint $k = N/2 = (L+1)/2$ of the chain would produce: $\phi_n(N/2) = \sin(n\pi/2)$. The substitution is justified by Part II of Conjecture 1, which posits equivalence of the resolvent-weighted limits for the true cascade profiles and the χ_4 effective model; §8 provides independent empirical support for this equivalence. Hence:

$$\text{Tr}[\mathcal{W} \cdot (I - \mathcal{K}_{\text{eff}})^{-1}] = \sum_{n=1}^{L} \frac{w_n}{1 - \lambda_n(A)} = \sum_{n \text{ odd}} \frac{\sin(n\pi/2)}{1 - \lambda_n(A)} \;=:\; S(A)$$

where the even-n terms vanish because $\sin(n\pi/2) = 0$ for even n, and $\lambda_n(A) = \frac{1}{2}\cos(n\pi/N) + A\sin^2(n\pi/(2N))$ for an L-site chain with Dirichlet walls at 0 and $N = L + 1$.

The bridge from discrete to continuous. Let $\beta = 1 - A$. Then:

$$\frac{1}{1 - \lambda_n} = \frac{2}{1 + \beta - \beta\cos(n\pi/N)} = \frac{2}{1+\beta} \cdot \frac{1}{1 - r\cos\theta_n}$$

where $r = \beta/(1+\beta)$ and $\theta_n = n\pi/N$.

The Poisson kernel expansion gives:

$$\frac{1}{1 - r\cos\theta} = \frac{1}{s}\left(1 + 2\sum_{m=1}^{\infty} \alpha^m \cos(m\theta)\right)$$

with $s = \sqrt{1-r^2}$, $\alpha = r/(1+s)$, $|\alpha| < 1$.

Substituting into the spectral sum and using the **Chebyshev orthogonality** relation (writing $N = L+1 = 2K$ for brevity; for K even, $N \equiv 0 \pmod 4$, and the mod-4 correction vanishes in the $K \to \infty$ limit regardless):

$$\sum_{k=0}^{N/2-1} (-1)^k \cos\big(m(2k+1)\pi/N\big) = \sec(m\pi/N) \text{ if } m \text{ is odd, and } 0 \text{ if } m \text{ is even.}$$

(where the odd-n sum with $\sin(n\pi/2) = (-1)^{(n-1)/2}$ has been reindexed as $n = 2k+1$), we obtain:

$$S = \frac{2}{1+\beta} \cdot \frac{2}{s} \sum_{m \text{ odd}} \frac{\alpha^m}{\cos(m\pi/N)}.$$

In the limit $L \to \infty$ (i.e., $N \to \infty$), $\cos(m\pi/N) \to 1$, giving:

$$S = \frac{4}{(1+\beta)\,s} \cdot \frac{\alpha}{1-\alpha^2} = \frac{4}{(1+\beta)\,s} \cdot \frac{r}{(1+s)^2 - r^2} \cdot \frac{1+s}{1}.$$

After simplification using $s^2 = 1 - r^2 = (1+2\beta)/(1+\beta)^2$ and $r = \beta/(1+\beta)$:

$$S = \frac{2\beta}{1+2\beta}. \qquad \text{(Phase Transition Formula)}$$

Where π enters. Setting $S = -\pi$:

$$\frac{2\beta}{1+2\beta} = -\pi \implies \beta = \frac{-\pi}{2+2\pi}, \implies A^* = 1 - \beta = \frac{2+3\pi}{2(1+\pi)}.$$

This is the *unique* value of A producing $S = -\pi$.

Remark (The role of χ_4 and the origin of π). The factor $\sin(n\pi/2) = \chi_4(n)$ is the non-principal Dirichlet character mod 4: it equals $+1, 0, -1, 0$ for $n = 1, 2, 3, 4$ (period 4). For odd n, it alternates as $+1, -1, +1, -1, \ldots$ for $n = 1, 3, 5, 7, \ldots$

The Chebyshev orthogonality (Step 2) is the key structural mechanism. After inserting the Poisson kernel into the χ_4-weighted sum, we must evaluate $\sum_{k=0}^{N/2-1} (-1)^k \cos(m\theta_k)$ with $\theta_k = (2k+1)\pi/N$. This sum vanishes for even m and equals $\sec(m\pi/N)$ for odd m. The χ_4 character therefore **projects the Fourier expansion onto its odd harmonics only**, producing the geometric series

$$\sum_{m \text{ odd}} \alpha^m = \alpha/(1-\alpha^2)$$

This is *thematically analogous to the Leibniz series* $1 - 1/3 + 1/5 - \cdots = \pi/4$: in both cases, selecting odd-indexed terms from an alternating series produces π. The mathematical mechanism here is different — a geometric series

$$\sum_{m \text{ odd}} \alpha^m = \alpha/(1-\alpha^2)$$

yielding a rational function of α, rather than the conditionally convergent harmonic alternation — but the structural principle is the same: a character projection onto odd modes extracts π from algebraic data. The standard (unweighted) resolvent integral involves all harmonics and yields an irrational answer containing $\sqrt{\pi+1}$; the χ_4 projection selects a subset whose alternating sum collapses to a rational function of β, and hence of π when $\beta = -\pi/(2+2\pi)$.

Remark. Writing $N = L+1 = 2K$, the Chebyshev orthogonality $\sum_{k=1}^{N-1} \sin(n\pi k/N)\sin(m\pi k/N) = (N/2)\delta_{nm}$ requires $N \equiv 0 \pmod 4$ (i.e., K even) to ensure that the character $\chi_4(n) = \sin(n\pi/2)$ is exactly representable on the lattice $\{k/N : k = 0, \dots, N-1\}$. For K odd ($N \equiv 2 \pmod 4$), a boundary correction of order $O(1/N)$ arises; this is $O(1/K)$ and vanishes in the sector-ratio limit $K \to \infty$.

Step 3: The Weierstrass Integral (Continuous Verification)

As an independent verification, consider the *unweighted* continuous resolvent trace:

$$I(z) = \int_0^\pi \frac{dk}{z - \frac{1}{2}\cos k}$$

with $z = -\frac{1}{2} - \frac{1}{\pi}$. The Weierstrass substitution $t = \tan(k/2)$, $\cos k = (1-t^2)/(1+t^2)$, $dk = 2\,dt/(1+t^2)$ gives the standard form:

$$\int_0^\pi \frac{dk}{a + b\cos k} = \frac{\operatorname{sgn}(a)\,\pi}{\sqrt{a^2-b^2}} \qquad (|a| > |b|)$$

with $a = z = -\frac{1}{2} - \frac{1}{\pi}$ and $b = -\frac{1}{2}$. The discriminant is:

$$a^2 - b^2 = \left(\frac{1}{2} + \frac{1}{\pi}\right)^2 - \frac{1}{4} = \frac{1}{4} + \frac{1}{\pi} + \frac{1}{\pi^2} - \frac{1}{4} = \frac{1}{\pi} + \frac{1}{\pi^2} = \frac{\pi+1}{\pi^2}$$

$$\sqrt{a^2-b^2} = \frac{\sqrt{\pi+1}}{\pi}$$

$$I(z) = -\frac{\pi^2}{\sqrt{\pi+1}}.$$

This integral computes $\sum_n 1/(z - \lambda_n^{(0)})$ over *all* modes, without the χ_4 weighting. Its value is irrational ($\sqrt{\pi + 1}$ appears). The weighted sum $S(A^*)$ selects only odd-n modes with alternating signs, and via the Poisson kernel derivation above, this selection rationalizes the answer to $S = 2\beta/(1 + 2\beta) = -\pi$.

The bridge between the two. The discrete χ_4-weighted sum and the continuous Weierstrass integral are related by the Poisson summation formula (or equivalently, the Chebyshev orthogonality identity). The unweighted integral encodes the *density of states* of the Markov operator; the χ_4 projection extracts the *odd-parity component* of this density, which is the component relevant to the D-odd boundary condition. The remarkable fact is that while the full density involves $\sqrt{\pi + 1}$, the odd-parity projection simplifies to a rational function of π. This simplification is structurally identical to the Leibniz phenomenon: $\sum 1/n^2 = \pi^2/6$ (all modes, irrational coefficient), while $\sum(-1)^{n+1}/n = \ln 2$ (alternating, no π-content) — the character projection transmutes the transcendental content. $\square$

---

8. Empirical Verification of Universal Mechanisms

This section presents the experimental evidence supporting Part II of Conjecture 1, and the systematic falsification of weaker hypotheses. Experiments E215–E219 use exact enumeration in Python for $K = 5, \dots, 15$; experiment E45 extends the per-position profile analysis to $K = 16, \dots, 21$ using a compiled C enumerator (§8.3, Table 3).

8.1 The Dirichlet Boundary Necessity (E218)

The D-odd condition (carry$[L] = 0$, where $L = 2K - 1$ as in §2.4) imposes Dirichlet boundary conditions on the carry chain. For D-even products (the carry chain extends one position further and carry$[L]$ is unconstrained), the carry chain is "full" — carries propagate to the top without interruption. Consequently:

- The cascade readout

$$\text{carries}[M_{\text{stop}} - 1] - 1 \equiv 0$$

for all D-even products. The D-even cascade is *degenerate*. - The schoolbook ratio converges to $R_0^{\text{even}} \approx +12.1$ — positive, purely logarithmic, and qualitatively different from $R_0^{\text{odd}} = -3.93$.

This proves that π is a property of the *Dirichlet boundary structure*, not of binary multiplication in general. Without the D-odd constraint, the sine eigenvector basis $\phi_n(k) = \sin(n\pi k/(L+1))$ does not exist, the spectral resolvent framework has no anchor, and the cascade degenerates.

8.2 Falsification of Parity Decay (E216)

If the universality mechanism were a simple "renormalization flow toward χ_4", the even-harmonic spectral energy should decay relative to odd-harmonic energy as K grows. Define:

$$\eta(K) = \frac{\sum_{n \text{ even}} |c_n|^2}{\sum_{n \text{ odd}} |c_n|^2}$$

where c_n are the Dirichlet sine coefficients of the cascade stopping profile. Experiment E216 measures $\eta(K)$ for $K = 5, \dots, 15$ across four profiles (stopping distribution, per-sector cascade, combined). Result:

$$\eta(K) \approx 1.000 \quad \text{for all } K.$$

No decay. The even and odd spectral energies are equal to three significant digits. The resolvent-weighted version $\eta_R(K)$ also converges to 1, not to 0. The "parity decay" universality hypothesis is *definitively falsified*.

Why parity is nonetheless irrelevant. The spatial integration weight $G_n = \sum_{d=1}^{N-1} \sin(n\pi d/N)$ vanishes identically for even n (a property of the sine function over a full half-period). Even modes do not contribute to the macroscopic observable

$$R = \sigma_{10,\text{tot}}/\sigma_{00,\text{tot}}$$

regardless of w_n. The falsification of parity decay does not threaten the spectral framework — it merely shows that the mechanism is *not* the naive suppression of even modes.

8.3 Resolvent Universality (E219, E45)

The decisive experiments project the per-sector cascade profiles onto the Dirichlet sine basis and reconstruct the resolvent sum with real spectral weights. Let $w_n^{(ab)} = \langle \sigma_{ab}, \phi_n \rangle$ be the n-th sine coefficient of the cascade profile for sector (a, b).

Key finding 1: Sectors are not spectrally proportional. The ratio $w_n^{(10)}/w_n^{(00)}$ varies wildly across modes (CoV > 1 for all K). The hypothesis that $\sigma_{10}(d) \propto \sigma_{00}(d)$ is *false*.

Key finding 2: The fundamental mode ratio overshoots $-\pi$ and diverges.

Table 3. Fundamental mode ratio, resolvent reconstruction, and raw $R(K)$.

K	$w_1^{(10)}/w_1^{(00)}$	gap$(n{=}1)$	$R_{\text{recon}} + \pi$	$R(K) + \pi$	|gap| ratio
10	-0.94	$+2.20$	$+2.000$	$+2.023$	$1.01\times$
12	-1.84	$+1.30$	$+1.144$	$+1.172$	$1.02\times$
14	-2.83	$+0.31$	$+0.488$	$+0.519$	$1.06\times$
15	-3.25	-0.11	$+0.285$	$+0.317$	$1.11\times$
16	-3.58	-0.44	$+0.155$	$+0.188$	$1.21\times$
17	-3.82	-0.68	$+0.074$	$+0.107$	$1.45\times$
18	-3.99	-0.85	$+0.026$	$+0.059$	$2.26\times$
19	-4.10	-0.96	-0.001	$+0.032$	$28\times$
20	-4.17	-1.03	-0.017	$+0.017$	$1.01\times$
21	-4.22	-1.08	-0.025	$+0.008$	$0.33\times$

The fundamental mode ratio crosses $-\pi$ between $K = 14$ and $K = 15$, then *continues to diverge* — reaching -4.22 at $K = 21$ (gap from $-\pi$: -1.08). The "Fundamental Mode Dominance" hypothesis is *definitively falsified*: the fundamental mode ratio diverges away from $-\pi$ for all $K \geq 15$.

Key finding 3: The resolvent reconstruction tracks $-\pi$ with oscillatory convergence. Computing $R_{\text{recon}} = S_{10}/S_{00}$ with the *actual* sine projections and LMH eigenvalues $\lambda_n(A^*)$ gives a gap from $-\pi$ that is dramatically smaller than the gap of $R(K)$ for $K \leq 19$. The resolvent gap decreases monotonically from $+2.00$ at $K = 10$ to $+0.026$ at $K = 18$, then *crosses zero* at $K \approx 19$, achieving an extraordinary near-hit: $|R_{\text{recon}} +$

$\pi| = 0.001$ at $K = 19$. Subsequently, R_{recon} overshoots $-\pi$, reaching a gap of -0.025 at $K = 21$.

This zero-crossing pattern — approach, near-hit, overshoot — is characteristic of finite-dimensional spectral truncation: the resolvent sum uses the first $L = 2K - 1$ modes, and the finite-L correction terms change sign as L increases. The overshoot amplitude (~ 0.025 at $K = 21$) is an order of magnitude smaller than the approach amplitude (~ 2.0 at $K = 10$), consistent with oscillatory convergence to $-\pi$ in the $L \to \infty$ limit. The raw $R(K)$, by contrast, approaches $-\pi$ monotonically from above.

Physical interpretation. The resolvent

$$(I - \mathcal{K}_{\text{eff}})^{-1}$$

, with its mode-dependent weights $G_n/(1 - \lambda_n)$, acts as a *matched spectral filter*: it combines information from all Fourier modes in proportions that are precisely tuned by the LMH eigenvalues. No single mode carries $-\pi$. Instead, the spectral structure of the entire operator encodes the transcendental constant through the collective weighting of modes whose individual ratios vary erratically. The near-hit at $K = 19$ ($|\text{gap}| = 0.001$, the closest any computed quantity reaches $-\pi$) demonstrates the power of the resolvent mechanism: the LMH eigenvalue structure transmutes the erratic cascade profiles into a macroscopic sum that oscillates tightly around $-\pi$.

8.4 Non-Triviality of the LMH (Observation 4 + Lemma 4)

Observation 4 (numerical). *For any diagonal cutoff function

$$\Omega = \text{diag}(\omega_1, \dots, \omega_L)$$

applied to a tridiagonal random walk K_M with eigenvalues $\cos(k\pi/(L+1))$:*

(a) (Trace obstruction.) $\text{Tr}(\Omega \cdot K_M) = 0$ while .

$$\text{Tr}(\mathcal{K}_{\text{eff}}) = LA^*/2 > 0$$

(b) The eigenvalue shifts are multiplicative, not additive.

$$\lambda_k^{\text{eff}} \approx \tilde{\Omega}_k \cdot \lambda_k^M,$$

inconsistent with the LMH form

$$\lambda_k = (1 - A)\lambda_k^M + A/2.$$

(c) Fitting A from slope vs. intercept yields a persistent discrepancy ≈ 0.57.

Evidence for (a)–(c): numerical computation for $L = 10, \dots, 500$ (E43). Part (a) trace = 0 follows from the zero-diagonal structure of the tridiagonal K_M, but the comparison with

$$\mathrm{Tr}(\mathcal{K}_{\text{eff}})$$

uses the LMH-predicted value A^*.

Lemma 4 (algebraic). The unique perturbation P satisfying the LMH spectral form $\lambda_k = (1-\varepsilon)\lambda_k^M + \varepsilon/2$ for all ε is $P = I - 2K_M$.

Proof. The LMH eigenvalue shift from λ_k^M to $(1-\varepsilon)\lambda_k^M + \varepsilon/2$ is $\varepsilon(1/2 - \lambda_k^M)$, giving the perturbation matrix $P = I/2 - K_M$ (up to the factor 2ε). In the tridiagonal basis: $P_{ij} = \delta_{ij}/2 - (K_M)_{ij}$, which is $I - 2K_M$ after normalization. □

The LMH requires the *specific* off-diagonal modification $\Delta K = -A(K_M - I/2)$, arising from bit-sharing correlations unique to binary multiplication.

8.5 Boundary Layer Decomposition (E223, E224b)

The cascade stopping distribution decomposes into universal per-depth contributions. Define $P(j \mid ab) = \lim_{K\to\infty} P(\text{cascade stops at depth } j \mid \text{sector } ab)$, where depth j means position $L - j$ from the top. Exact enumeration (E223, $K = 5, \dots, 12$; E224b, $K = 7, \dots, 12$) establishes convergence:

Table 4. Universal cascade stopping probabilities $P(j \mid \text{sector})$. Values from $K = 12$ exact enumeration (convergence to $< 1\%$ relative for $j \le 7$). Updated 5-digit-precision values from $K = 21$ exact enumeration (3.4×10^{11} D-odd pairs in the $X \le Y$ half-plane; full count $\approx 6.8 \times 10^{11}$) are in [P1, §8.2a, Table 1].

j	$P(j \mid 00)$	$P(j \mid 10)$	Comment
2	0.544	**0**	Structural exclusion
3	0.138	0.460	Dominant sector-10 depth
4	0.110	0.230	
5	0.086	0.145	
6	0.050	0.079	
7	0.033	0.043	
8	0.018	0.023	

Structural exclusion at $j = 2$. The exclusion $P(2 \mid 10) = 0$ is a rigorous consequence of the carry recurrence. At position $L - 2$, the convolution is

$$\mathrm{conv}[L-2] = a_1 b_0 + a_0 b_1,$$

where $a_0 = b_0 = 1$ (leading bits, writing b_j for the second operand's digits as distinct from the sector label c_1 of §2.3). In sector $(1, 0)$: $a_1 = 1, c_1 = 0$ (i.e., $b_1 = 0$), so

$$\text{conv}[L-2] = 1.$$

The carry constraint $\text{carry}[L-1] = 0$ then forces

$$\lfloor(\text{carry}[L-2] + 1)/2\rfloor = 0,$$

hence $\text{carry}[L-2] = 0$ — no stopping event is possible. In sector $(0,0)$:

$$\text{conv}[L-2] = 0,$$

so

$$\lfloor\text{carry}[L-2]/2\rfloor = 0$$

allows $\text{carry}[L-2] \in \{0,1\}$.

Sector $(1,1)$ is D-odd impossible. With $a_1 = b_1 = 1$: $\text{conv}[L-2] = 2$, and $\lfloor(c+2)/2\rfloor = 0$ has no solution for $c \geq 0$.

Global entanglement. The backward-tree DP (E224b), which starts from $\text{carry}[L] = 0$ and propagates downward assuming uniform product-bit probabilities, recovers the structural exclusions exactly but produces quantitatively incorrect stopping probabilities (e.g.,

$$P_{\text{tree}}(2 \mid 00) = 0.775$$

vs. exact 0.544, a 42% relative error). This demonstrates that the D-odd boundary condition creates correlations between the boundary layer and the bulk carry distribution that cannot be captured by local approximations.

---

9. Conclusion and Open Problems

9.1 Summary

We have established a conditional spectral framework together with strong numerical evidence consistent with the convergence of the sector ratio $R(K)$ to $-\pi$, and shown that:

1. The schoolbook component R_0 is exact and logarithmic (Proposition 1).
2. The transcendental correction ΔR is governed by a resolvent trace whose spectral structure, under both parts of the LMH (spectral structure + resolvent universality), yields $-\pi$ exactly (Theorem 1, conditional).
3. The emergence of π requires the Dirichlet boundary structure imposed by D-odd parity; D-even products yield a degenerate cascade and a purely logarithmic schoolbook limit (§8.1).
4. The true universality mechanism is *resolvent universality*: the resolvent-weighted spectral sum of the actual cascade profiles tracks $-\pi$ with oscillatory convergence, achieving a near-hit of $|R_{\text{recon}} + \pi| = 0.001$ at $K = 19$ (§8.3, Table 3). The effective spectral parameter $A_{\text{eff}}(K)$ converges to A^* with Aitken-extrapolated accuracy of 0.005% at $K = 21$.
5. The LMH is non-trivial: no generic cutoff operator produces it (Observation 4 + Lemma 4, §8.4).
6. The microscopic mechanism is identified: the D-odd constraint imposes rigid structural exclusions on the boundary carry positions — in particular, cascade stopping at depth $j = 2$ is impossible in sector 10, and sector 11 is entirely excluded. These structural exclusions are proven analytically via the backward-tree DP (§8.5); the per-depth stopping probabilities $P(j \mid \text{sector})$ are measured by exact enumeration (Table 4) and converge rapidly with K.
7. The transcendental constant π resides in the *cascade dynamics*, not in the counting geometry: the sector count ratio n_{00}/n_{10} converges to an algebraic constant ≈ 3.126, not to π (§9.2).

9.2 The Trace Anomaly is Dynamical, Not Geometric

The sector ratio decomposes as $R(K) = \bar{v}_{10}(K)/(\ell(K) \cdot \bar{v}_{00}(K))$, where $\bar{v}_{ab}$ is the mean cascade value in sector ab and $\ell(K) = n_{00}(K)/n_{10}(K)$ is the cascade count ratio. Aitken extrapolation on the count ratio yields

$$\ell(K) \longrightarrow L_\infty \approx 3.1257 \qquad (K \to \infty)$$

which is close to $25/8 = 3.1250$ but emphatically *not* $\pi = 3.1416\ldots$ — the gap $|L_\infty - \pi| \approx 0.016$ is four orders of magnitude larger than the exponential tail $\sim 2^{-K}$ expected for quantities converging at the Diaconis–Fulman rate. The count ratio is an algebraic (likely rational) constant determined by the geometric phase-space volumes of the sectors under D-odd conditioning.

The transcendental content of the conjectured limit $R \to -\pi$ therefore resides entirely in the *cascade values*, not in the sector counting measure. The mean-value ratio $\bar{v}_{10}/\bar{v}_{00}$ must compensate the algebraic ℓ to produce $-\pi$: this compensation is

the trace anomaly in its purest form. If the LMH holds (Conjecture 1), the LMH operator — not the geometry of the D-odd domain — is the mechanism that encodes π into the cascade dynamics.

9.3 Open Problems

Open Problem 1 (Derivation of A^* via the Holte Bridge)**.** *Derive the spectral parameter*

$$A^* = \frac{2 + 3\pi}{2(1 + \pi)}$$

from the combinatorial structure of the D-odd carry chain.

The microscopic mechanism is now understood (§8.5): the D-odd constraint carry$[L] = 0$ propagates backward through the carry chain, creating rigid structural exclusions (sector $(1, 0)$ blocked at depth $j = 2$; Table 4) and universal per-depth stopping constants. The remaining step is to evaluate the Doob h-transform (bridge process) of the Holte–Diaconis–Fulman Markov chain conditioned on carry$[L] = 0$, i.e., the stationary distribution $\pi_{\text{bridge}}(c)$ of the carry entering the boundary layer. Specifically:

1. *Holte bridge:* Compute the carry distribution at position $L - j$ for the Markov chain on $\{0, 1, ...\}$ started from carry$[0] = 0$ and conditioned to return to 0 at position L, in the limit $L \to \infty$.
2. *Boundary filter:* Propagate $\pi_{\text{bridge}}(c)$ through the exact exclusion operators of §8.5 (the backward-tree DP of E224b), obtaining the universal stopping probabilities $P(j \mid \text{sector})$ and expected values

$$E[\text{val} \mid j, \text{sector}]$$

analytically. 3. *Spectral closure:* Show that the resulting weighted sum of per-depth cascade contributions yields $\Delta R = |R_0| - \pi$, thereby determining A^*.

The local approximation $P(\text{bit}) = 1/2$ (E224b) captures the structural exclusions exactly but introduces a $\sim 23\%$ quantitative error, demonstrating that the bridge distribution carries essential global information from the D-odd constraint.

Open Problem 2 (Resolvent Universality and the Gibbs Phenomenon)**.** *Prove that the resolvent-weighted spectral sum*

$$R_{\text{recon}} = \frac{\sum_n w_n^{(10)} G_n/(1 - \lambda_n(A^*))}{\sum_n w_n^{(00)} G_n/(1 - \lambda_n(A^*))} \longrightarrow -\pi$$

in the limit $L \to \infty$, where $w_n^{(ab)}$ are the Dirichlet sine projections of the true per-sector cascade profiles.

Table 3 establishes the empirical picture: (i) the full spectral projection w_n is *not* proportional to χ_4 at any finite K (E215); (ii) even-harmonic energy does not decay (E216, $\eta \approx 1$); (iii) the mode-by-mode ratio $w_n^{(10)}/w_n^{(00)}$ varies wildly across modes (E219); (iv) the fundamental-mode ratio $w_1^{(10)}/w_1^{(00)}$ overshoots $-\pi$ at $K \geq 15$ and diverges (reaching -4.22 at $K = 21$); but (v) the resolvent reconstruction R_{recon} tracks $-\pi$ with oscillatory convergence, achieving $|R_{\text{recon}} + \pi| = 0.001$ at $K = 19$ before overshooting; and (vi) the cascade stopping distribution decomposes into universal per-depth probabilities with a rigid structural exclusion at depth $j = 2$ in sector $(1, 0)$ (§8.5, Table 4).

The oscillatory pattern at $K \geq 19$ — approach, near-hit, overshoot — is the spectral analogue of the **Gibbs phenomenon**: the finite-L resolvent truncation overshoots the true limit $-\pi$ because higher-order correction modes change sign as L increases. This is *stronger* evidence for a spectral sum than monotone convergence would be, because Gibbs-type oscillation is the signature of a truncated Fourier/resolvent series, not of a simple Markov probability decay.

Open Problem 3 (Extension to general s). *Determine the analytic form of the coefficient $B(s)$ in the critical-line decomposition*

$$\mu_{\chi_4}(s) \approx c_{\text{carry}} \cdot L(s, \chi_4) + B(s) \cdot L'(s, \chi_4),$$

where $c_{\text{carry}} = \lim_{\sigma\to\infty} \tilde{c}(\sigma) \approx 0.0556 \approx 1/18$ *and* $B(1/2 + it) \sim 0.041 \cdot t^{-0.23}$ *with* $B(\infty) \approx 0.012$. Paper [L] extends the carry–χ_4 channel from a single scalar value to a meromorphic function of s on all of $\mathrm{Re}(s) > 0$ (see [L, §7.4]). The spectral parameter A^* of the LMH manifests in this extension through the structure of the zeros of μ_{χ_4} on the critical line: the positions of the minima of $|\mu_{\chi_4}(1/2+it)|$ coincide with the zeros of $L(s, \chi_4)$ and are topologically stable under $K = 21 \to K = 999$ ([L, §5.5]; [P1, §9.5]). The link between A^* and the analytic form of $B(s)$ — for instance, whether $B(s)$ arises from the resolvent $(I - T_{A^*})^{-1}$ restricted to the derivative sector — remains open.

9.4 Broader Perspective

The emergence of π from carry arithmetic — a process defined entirely in terms of integer floor functions and binary digit manipulations — illustrates a general principle: transcendental constants can arise as *spectral invariants* of combinatorial operators, even when every finite-order computation yields only algebraic (logarithmic) quantities. The carry chain provides a concrete, computable instance of this phenomenon.

The falsification of the geometric hypothesis ($n_{00}/n_{10} \not\to \pi$, §9.2) sharpens this message: π is not hiding in the combinatorial geometry of the D-odd domain. It is a **dynamical invariant** — a property of the spectral transfer operator acting on the cascade, mediated by the Dirichlet boundary condition. The "trace anomaly" is precisely the gap between the algebraic counting measure and the transcendental spectral weight.

Appendix A: Numerical Data

A.1 Exact $R(K)$ Values

Table 1. Exact sector ratio $R(K)$ for $K = 4$ to 21 (exhaustive enumeration; E44 for $K \leq 20$, E45 for $K = 21$). The cascade valuation counts only pairs where the cascade stops (first nonzero carry from top); pairs with no nonzero carry are excluded. All values are exact integer ratios; the table shows 6-digit decimal approximations. The column $g(K)/g(K-1)$ is the consecutive gap ratio, converging toward $1/2$ (the Diaconis–Fulman spectral radius).

K	$R(K)$	$g(K) = R(K) + \pi$	$g(K)/g(K-1)$
4	$+0.2500$	$+3.392$	—
5	$+0.2500$	$+3.392$	1.000
6	$+0.0930$	$+3.235$	0.954
7	-0.0915	$+3.050$	0.943
8	-0.3848	$+2.757$	0.904
9	-0.7262	$+2.415$	0.876
10	-1.1184	$+2.023$	0.838
11	-1.5535	$+1.588$	0.785
12	-1.9696	$+1.172$	0.738
13	-2.3388	$+0.803$	0.685
14	-2.6225	$+0.519$	0.647
15	-2.8241	$+0.317$	0.612
16	-2.9539	$+0.188$	0.591
17	-3.0348	$+0.107$	0.569
18	-3.0822	$+0.059$	0.557
19	-3.1095	$+0.032$	0.540
20	-3.1249	$+0.017$	0.522
21	-3.1334	$+0.0082$	0.491

On the strength of the convergence claim. At $K = 21$ the raw gap is $g = 0.0082$ (2.1 significant digits of π). The gap ratios form a monotone sequence converging to $1/2$ from above: $1.000, 0.954, \ldots, 0.522, 0.491$. Three independent lines of evidence provide much stronger confirmation than the raw table alone:

1. **The continuum cancellation series** (§4.2) operates in the $K \to \infty$ limit directly. The partial sum $\sum_{d=1}^{14} f(d) = -8.7 \times 10^{-5}$, with subsequent terms decaying as $\sim C/2^d$, constrains $|\sum_{d=1}^{\infty} f(d)| < 10^{-4}$, corresponding to $|R+\pi| < 10^{-4}$ in the continuum.

2. **Two-point convergence scaling** (Appendix C) applied to $R(K)$ at $K = 20, 21$ yields $R_\infty \approx -3.14$; Aitken Δ^2 acceleration on $f(d)$ gives $\sum f \approx 7 \times 10^{-6}$. The identification $R = -\pi$ is not based on the raw $R(21) = -3.133$ alone.

3. The A_{eff} convergence (§6.1, Appendix A.2) provides an internal consistency check of the LMH ansatz: the spectral parameter extracted from $R(K)$ converges to $A^* = (2 + 3\pi)/(2(1 + \pi))$ with Aitken-extrapolated value 1.37934

(from $K = 19, 20, 21$), within 0.005% of $A^* = 1.37927$.

A.2 Effective Spectral Parameter

Table 2. Effective spectral parameter $A_{\text{eff}}(K)$ extracted from $R(K)$ via inversion of $S(A) = 2(1-A)/(3-2A)$. Convergence target: $A^* = (2+3\pi)/(2(1+\pi)) = 1.37927\ldots$.

K	$A_{\text{eff}}(K)$	$A^* - A_{\text{eff}}$	gap ratio
4	0.833	0.546	—
5	0.833	0.546	1.00
6	0.949	0.431	1.27
7	1.042	0.337	1.28
8	1.139	0.240	1.40
9	1.210	0.169	1.42
10	1.264	0.115	1.47
11	1.304	0.075	1.54
12	1.332	0.048	1.58
13	1.350	0.029	1.64
14	1.362	0.017	1.68
15	1.369	0.010	1.73
16	1.374	0.0057	1.75
17	1.376	0.0032	1.79
18	1.378	0.0018	1.82
19	1.378	0.00094	1.87
20	1.379	0.00049	1.92
21	1.3790	0.00024	2.04

The gap ratio increases monotonically toward 2, consistent with $\delta(K) \sim C \cdot 2^{-K}$ for large K (exponential convergence at rate $\rho = 1/2$). At $K = 21$ the gap ratio crosses 2 for the first time, confirming the asymptotic regime. Aitken Δ^2 acceleration on the last three values ($K = 19, 20, 21$) gives

$$A_{\text{eff}}^{(\infty)} = 1.37934$$

, compared to $A^* = 1.37927$ — a relative error of 0.005%.

A.3 Key Constants

$$R_0 = \frac{\frac{1}{2} - 2\ln\frac{4}{3}}{1 + \ln\frac{3}{8}} = -3.931205442837\ldots$$

$$A^* = \frac{2+3\pi}{2(1+\pi)} = 1.379273496536\ldots$$

$$z^* = -\frac{1}{2} - \frac{1}{\pi} = -0.818309886184\ldots$$

$$1 - A^* = \frac{-\pi}{2(1+\pi)} = -0.379273496536\ldots$$

Appendix B: The Continuum DFS Algorithm

The per-depth quantities $\sigma_{ij}(d)$ are computed by a depth-first search over dyadic cells. At depth d, the cell is specified by the bit sequences $(a_1, \dots, a_d)$, $(c_1, \dots, c_d)$, and $(b_1, \dots, b_d)$, which determine:

- The u-interval $[u_{\text{lo}}, u_{\text{hi}}]$ and v-interval $[v_{\text{lo}}, v_{\text{hi}}]$ (dyadic subintervals of $[0, 1]$)
- The t-strip $[t_{\text{lo}}, t_{\text{hi}})$ where $t = (1 + u)(1 + v)$ ranges over the product constraint
- The carry state at each position

The D-odd constraint requires $t < 2$, so the effective v-upper bound is

$$v_{\text{max}}(u) = \min(v_{\text{hi}}, (t_{\text{hi}} - 1 - u)/(1 + u)).$$

The area integral

$$\int_{u_{\text{lo}}}^{u_{\text{hi}}} [v_{\text{max}}(u) - v_{\text{lo}}]\, du$$

is computed exactly using partial fractions, yielding rational combinations of logarithms of rationals. The exact computation uses `mpmath` arbitrary-precision arithmetic (50 decimal digits).

Appendix C: Series Acceleration

The gap $g(K) = R(K) + \pi$ decays with a rate that accelerates toward $\rho = 1/2$ (the Diaconis–Fulman spectral radius). The consecutive gap ratio $g(K)/g(K-1)$ decreases monotonically: $1.000, 0.954, 0.943, \ldots, 0.540, 0.522$ (Appendix A.1), approaching $1/2$ from above.

Two-point convergence scaling of $R(K)$. Assuming geometric convergence $g(K) \sim C\rho^K$ with $\rho \approx g(K)/g(K-1)$, we estimate the limit from two consecutive values via $R_\infty = R(K_2) - g(K_2)/(1-\rho)$ where $\rho = g(K_2)/g(K_2-1)$. Taking $(K_2, \rho) = (20, 0.522)$:

$$R_\infty = -3.1249 - \frac{0.0167}{1-0.522} = -3.1249 - 0.035 = -3.16 \quad \text{(crude; see Aitken below)}.$$

The slow convergence of ρ toward $1/2$ limits the precision of two-point extrapolation. The Aitken method below is far more effective.

Aitken Δ^2 acceleration of $A_{\text{eff}}(K)$. This provides a more sensitive test because A_{eff} is a nonlinear transform of R that amplifies the asymptotic signal. Applied to $(K = 19, 20, 21)$:

$$A_{\text{eff}}^{(\infty)} = A_{\text{eff}}(19) - \frac{(A_{\text{eff}}(20) - A_{\text{eff}}(19))^2}{A_{\text{eff}}(21) - 2\,A_{\text{eff}}(20) + A_{\text{eff}}(19)} = 1.37934,$$

compared to the predicted $A^* = 1.37927$. The relative error is 0.005%. The Aitken values over the triple sequence converge monotonically: $1.390, 1.385, 1.381, 1.380, 1.380, 1.3797, 1.37$ for $(K, K+1, K+2) = (11, 12, 13)$ through $(19, 20, 21)$, all approaching A^* from above.

Aitken Δ^2 on the cancellation series $f(d)$ (§4.2) yields $\sum f \approx 7 \times 10^{-6}$, consistent with convergence to zero.

References

External

1. J. M. Holte, *Carries, combinatorics, and an amazing matrix*, American Mathematical Monthly **104** (1997), 138–149.
2. P. Diaconis and J. Fulman, *Carries, shuffling, and symmetric functions*, Advances in Applied Mathematics **43** (2009), 176–196.

Companion Papers (this series)

3. [A] Companion paper: *Spectral Theory of Carries in Positional Multiplication*. (Foundation: the m-bit Equidistribution Lemma extending Diaconis–Fulman; eigenvalue structure used throughout §2.5, §6, §7.). doi:10.5281/zenodo.18895593
4. [B] Companion paper: *Carry Polynomials and the Euler Product: An Approximation Framework*. (Central approximation framework; the Carry Representation Theorem and trace anomaly coefficient $c_1 = \pi/18$.). doi:10.5281/zenodo.18895597
5. [F] Companion paper: *Exact Covariance Structure of Binary Carry Chains*. (Exact second-order structure; the bit-sharing correlations underlying the LMH perturbation §6.3.). doi:10.5281/zenodo.18895607
6. [P1] Companion paper: *π from Pure Arithmetic: A Spectral Phase Transition in the Binary Carry Bridge*. (The spectral phase-transition narrative; introduces the conjecture $R(K) \to -\pi$ and the Carry Bridge framework.). doi:10.5281/zenodo.18895611
7. [P2] Companion paper: *The Sector Ratio in Binary Multiplication: From Markov Failure to Transcendence*. (Defines the sector ratio $R(K)$; provides the exact enumeration data and the Markov-failure structural decomposition used in §3 and Appendix A.). doi:10.5281/zenodo.18895615
8. [G] Companion paper: *The Angular Uniqueness of Base 2 in Positional Multiplication*. (Proves that the straight D-parity boundary $\alpha+\beta = \pi/4$ is unique to base 2; the angular coordinates of §2.2 originate here.). doi:10.5281/zenodo.18895601